

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 2: Polynomials - Detailed Guide

CHAPTER 2: POLYNOMIALS

A simplified, detailed guide in Hinglish

For students who missed Class 7-8-9

Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION (Algebra ki ABCD)

1.1 VARIABLE KYA HOTA HAI?

VARIABLE = ek alphabet (jaise x, y, z) jo kisi UNKNOWN number ko represent karta hai.

EXAMPLE:

Maan le tere paas kuch chocolates hain, par tu count nahi jaanta. Tu kahega "x chocolates hain mere paas". Yahan x variable hai.

1.2 CONSTANT KYA HOTA HAI?

CONSTANT = jiska value FIXED ho.
Numbers like 5, -3, $\frac{1}{2}$, pi - sab constant hain.

1.3 ALGEBRAIC EXPRESSION

EXPRESSION = variable + constant + operations
(+, -, x, /) ka mixture.

EXAMPLES:

- $2x + 5$ (variable x, constants 2 and 5)
- $3x^2 - 7x + 1$
- $5y + 3$

1.4 POLYNOMIAL - ASLI DEFINITION

POLYNOMIAL = ek special type ka algebraic expression jisme:
- Variable ke power WHOLE NUMBERS (0,1,2,3...) ho
- Power NEGATIVE ya FRACTION nahi ho

POLYNOMIAL HAI:

- $2x + 3$ (power 1)
- $x^2 - 5x + 6$ (power 2)
- 7 (constant, power 0)

POLYNOMIAL NAHI HAI:

- $x^{-2} + 3$ (negative power)
- $\sqrt{x} + 1$ ($= x^{1/2}$, fraction power)
- $1/x$ ($= x^{-1}$, negative power)

1.5 DEGREE OF POLYNOMIAL

DEGREE = polynomial mein HIGHEST power of variable.

EXAMPLES:

Polynomial	Highest Power	Degree
-----	-----	-----
$2x + 5$	1	1
$3x^2 - 7x + 1$	2	2
$x^3 + 2x^2 - x + 4$	3	3
7	0	0

1.6 TYPES OF POLYNOMIALS (degree ke hisab se)

Degree	Naam	Form	Example
-----	-----	-----	-----
0	Constant Polynomial	c	5, -3, 7
1	LINEAR Polynomial	$ax + b$	$2x + 3$
2	QUADRATIC Polynomial	$ax^2 + bx + c$	$x^2 - 5x + 6$
3	CUBIC Polynomial	$ax^3 + bx^2 + cx + d$	$x^3 - 2x + 1$

IMPORTANT: $a \neq 0$ hona chahiye (warna degree change ho jayegi).

1.7 ZERO / ROOT OF A POLYNOMIAL

ZERO = wo value of x jiske liye polynomial ki value 0 ho jaye.

EASY WAY TO FIND:

$p(x) = 0$ set karo aur x solve karo.

EXAMPLE: $p(x) = 2x - 6$ ka zero nikaal

$$2x - 6 = 0$$

$$2x = 6$$

$$x = 3$$

So zero = 3.

CHECK: $p(3) = 2(3) - 6 = 6 - 6 = 0$. Sahi!

Q AND A TIME - SECTION 1 KA TEST

Q1. In mein se POLYNOMIAL konse hain? (Yes/No)

(a) $3x^2 + 5x - 2$

(b) $2x^{-1} + 4$

(c) $\sqrt{x} + 3$

(d) $7x^3 - x + 1$

(e) 5

Q2. Inka DEGREE bata:

(a) $4x^5 + 3x^2 - 1$

(b) $7x + 2$

(c) 9

Q3. $p(x) = 5x^3 - 2x^2 + 7x - 1$ polynomial ka type kya hai?

(Linear / Quadratic / Cubic)

Q4. $p(x) = 3x + 9$ ka zero nikaal.

Q5. Agar $p(x) = x^2 - 4$, toh:

(a) $p(2)$ ki value kya hai?

(b) $p(-2)$ ki value kya hai?

(c) Inn jawabon se kya conclusion nikalta hai?

PROGRESS STATUS

[DONE] Section 1: Foundation

[TO DO] Section 2: Geometrical Meaning of Zeroes

[TO DO] Section 3: Relationship between Zeroes
and Coefficients

[TO DO] Practice Problems

[TO DO] Summary

(Aage ke sections aate jayenge as we progress
chapter mein.)

Generated by Kiro for Krishna1k
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